

**DEPARTMENT OF PHYSICS
UNIVERSITY OF PERADENIYA**

Laboratory Courses

Error Theory

Introduction

In Physics, we do predictions, make hypotheses, and use laws and theories to understand the physical world around us. In addition to that, we do experiments and measurements to test hypotheses and verify laws and theories. Physical quantities cannot be measured exactly. Always there is an error associated with any measurement. When we take measurements, we always try to make the error minimum. There is no measurement with zero error.

There are few misconceptions about error.

- Error is the difference between the measured value and its standard value.
- Standard values have no errors associated.

The error is not the difference between the measured value and its standard value. Also, all the standard values of physical quantities are also found using experiments, and therefore standard values have errors.

Error or Uncertainty

Error is the difference between the actual value (impossible to measure) and the measured value of a physical quantity. The actual value of a physical quantity cannot be measured; therefore, the actual error of a measurement cannot be found. Therefore, we do error estimation or prediction. Error of a measurement of a physical quantity will be estimated by considering the measuring tools and the techniques used.

Physics is a quantitative science and always deal with numbers which are the measurements of physical quantities. A quantitative measurement consists of a numerical quantity, appropriate unit and the error or the uncertainty associated with the measurement.

Example: $(5.45 \pm 0.01) \times 10^{-1} \text{ m}$ or $(54.5 \pm 0.1) \text{ cm}$

When stating a physical measurement, we need to consider about the significant figures and decimal places of the numerical value of the measurement.

Significant Figures

It is important to be honest when reporting a measurement, so that it does not appear to be more accurate than the accuracy of the equipment used. We can achieve this by controlling the number of digits, or significant figures, used to report the measurement. The significant figures of a number are the digits that are used to express the number to the required degree of accuracy.

Example: Consider a measurement of the length of a pencil.

If the length is measured with a meter ruler, the length would be 17.4 cm and if the measurement of the length is made with a Vernier caliper, the length would be 17.43 cm. 17.4 cm has 3 significant figures and 17.43 cm has 4 significant figures. Therefore, 17.43 cm is more accurate than 17.4 cm.

There are three rules on determining how many significant figures are in a number:

Rule 1: The leftmost non-zero digit is the most significant.

Rule 2: If there is no decimal point, the rightmost non-zero digit is the least significant. If there is a decimal point, the rightmost digit is the least significant even if it is zero.

Rule 3: All the digits from most significant to least significant are significant.

Example: In the number 3060700, 3 is the most significant and 7 is the least significant, therefore it has five significant figures.

But trailing zeros in 3060700 may or may not be significant, depending on the measurement or reporting resolution. 3060700 has 5, 6 or 7 significant figures depending on how the last zeros are determined. For example, if a length is reported as 3060700 cm without information about the reporting or measurement resolution, then it is not clear if the length is precisely measured as 3060700 cm or if it is a rough estimate. If it is the rough estimation, then only the first five digits are significant since the trailing zeros are not reliable; 3060700 cm can be expressed as 30607 m or 30.607 km or as 3.0607×10^6 cm in scientific notation, and none of the expressions require the trailing zeros.

In the number 11.020, the leftmost 1 is the most significant and the rightmost zero is the least significant, therefore it has five significant figures.

In the scientific notation, all the significant figures of the number are used.

Examples:	201.3		2.013×10^2
	0.0612		6.12×10^{-2}
	300000		3×10^5
	25.70		2.570×10^1
	0.0031500		3.1500×10^{-3}

Mathematical calculations with Significant figures

In mathematical operations involving significant figures, the answer is reported in such a way that it reflects the reliability of the least precise operation. An answer is no more precise than the least precise number used to get the answer.

Consider the multiplication of two numbers: 23.4×18.2 .

The mathematical result of the multiplication 23.4×18.2 is 425.88. But physically this is not the correct result, because both numbers are numbers with three significant figures. Therefore, the next decimal place of the number is unknown.

Consider the following example.

$$\begin{array}{r}
 23.4? \text{ cm (3 s.f.)} \\
 \times 18.2? \text{ cm (3 s.f.)} \\
 \hline
 \quad ??? \\
 468? \\
 1872? \\
 234? \\
 \hline
 \underline{\underline{425.???? \text{ cm}^2}}
 \end{array}$$

When we do the multiplication with the unknown decimal place as above, the final answer becomes 425 with unknown decimal places. Therefore, when multiplying (or dividing) numbers, the result should be expressed keeping only the number of significant figures of the least accurate number used in the mathematical operation of multiplication or division. Therefore, the final answer should be expressed with three significant figures as 426 after rounding the value (425.88) calculated using the calculator.

Now consider the addition of 88.932 and 4.24. The mathematical sum is 93.172. But since the 3rd decimal place in 4.24 is unknown, the physical result of the addition will be 93.17 as shown below.

$$88.932 + 4.24? = 93.17?$$

When adding or subtracting numbers, the result should be expressed with the same number of decimal places as of the number with the least number of decimal places.

Now consider taking the **logarithm of a number**. The number of digits in the decimal part of the logarithm of a number are the only significant figures in the logarithm of a number. The number of decimal places of the logarithm of a number should be the number of significant figures of the number in which the logarithm is taken. The integer, or characteristic, is the power to which the base (in the following examples, 10) is raised when writing the number in scientific notation. It is an exponent and therefore not a significant figure.

Examples:

$$\begin{aligned}
 \text{(i). } 24.3 &= 2.43 \times 10^1 \rightarrow \log(2.43 \times 10^1) = 1.3856 \rightarrow \log(24.3) = 1.386 \\
 \text{(ii). } 0.068 &= 6.8 \times 10^{-2} \rightarrow \log(6.8 \times 10^{-2}) = -1.1674 \rightarrow \log(0.068) = -1.17
 \end{aligned}$$

The decimal part in above examples have the same number of digits as the number of significant figures as the number from which it was derived. When calculating natural logarithms (base e), the same rules apply. The above examples yield:

$$\begin{aligned}
 \text{(i). } \ln(24.3) &= 3.1904 \rightarrow \ln(24.3) = 3.190 \\
 \text{(ii). } \ln(0.068) &= -2.6882 \rightarrow \ln(0.068) = -2.69
 \end{aligned}$$

Exponentiation or conversion of logarithms back into their corresponding numbers (called antilog) uses the same rules backwards. The number of digits in the decimal part gives the number of significant figures in the resulting number.

Examples:

$$\text{(i). } \text{antilog}(23.32) = 10^{23.32} = 2.0892 \times 10^{23} \rightarrow \text{antilog}(23.32) = 2.1 \times 10^{23}$$

- (ii). $\text{antilog}(-1.873) = 10^{-1.873} = 0.013396 \rightarrow \text{antilog}(-1.873) = 0.0134$
- (iii). $e^{23.32} = 1.3419 \times 10^{10} \rightarrow e^{23.32} = 1.3 \times 10^{10}$
- (iv). $e^{-1.873} = 0.15366 \rightarrow e^{-1.873} = 0.154$

Angles in Degrees and Radians

Most of the time, we measure angles in degrees. Calculations are sometimes done using radians. In most of the measurements of angles (ex: using spectrometer), angles can be measured with a precision of 1 minute.

The number of significant figures of an angle in radians should be the number of significant figures of the angle experimentally measured in degrees.

Example: One can measure an angle using the spectrometer to be $38^\circ 46'$ which has four significant figures. Therefore, the angle in radians also should have four significant figures.

In radians, the angle should be $\pi \times 38^\circ 46' / 180^\circ 00' = 0.6767$

Rounding Numbers

It is often necessary to drop one or more digits to obtain an answer with the appropriate number of significant figures. The rules are:

- If the first digit to be dropped is less than 5, leave the preceding digit unchanged. (e.g., 3.12467 rounds to 3.12 with three significant figures because the first number to be dropped is 4.)
- If the first digit to be dropped is greater than or equal to 5, increase the preceding digit by 1. (e.g., 4.26713 rounds to 4.27 with three significant figures because the first number to be dropped is 7.)

Error and Uncertainty

No measurement of a continuous physical variable is exact. Some degree of uncertainty or error is associated with the measured data of all experiments.

There are three types of limitations to measurements:

Instrumental limitations: Any measuring device can only be used to measure with a certain degree of accuracy. Accuracy of our measurements cannot be better than the accuracy of the instruments we use to produce them.

Systematic errors and blunders: These are caused by a mistake which does not change during the measurement. For example, if the platform of a balance you used to weigh something was not correctly set to zero with no weight on the pan, all your subsequent measurements of mass would be too large or too small. Systematic errors do not enter into the uncertainty. They are either identified and eliminated or lurk in the background producing a shift from the true value.

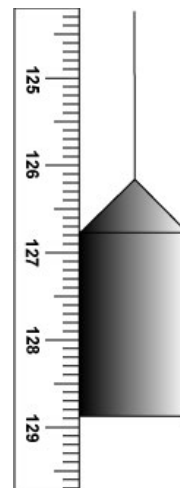
Random errors: These arise from unavoidable variations or changes in the experimental environment, etc. Random variations affect precision. Truly random effects average out if the results of a large number of trials are combined.

Errors Associated with Scale Readings

The most common measurements in the lab are done with devices that have a marked scale like a meter ruler.

Let us look at an example. In one of your first experiments, you will measure the length of a pendulum from the pivot point to the center of the mass attached to its end point. We will measure the length of the pendulum from the pivot point to the visible end of the mass. The situation is schematically shown in the figure below.

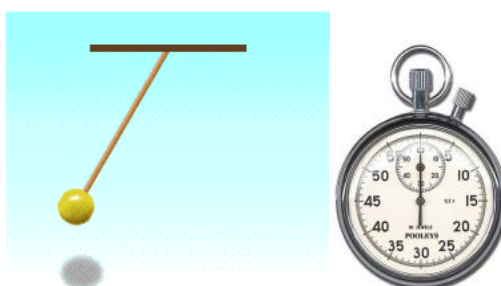
Even using this idealized, zoomed-in picture we cannot tell for sure whether the length to the end of the mass is 128.89 cm or 128.88 cm. However, it is certainly closer to 128.9 cm than to 128.8 cm or 129.0 cm. Thus, we can state with absolute confidence that the length L is 128.9 ± 0.1 cm. We call the first term, 128.9 cm, the "central value" and the second term, 0.1 cm, the "error" or "uncertainty". If pressed, we could get a little bit better precision from the picture. However, in a real situation, the precision of 0.1 cm for measurements done with the meter ruler is as good as you can get.



Statistical Error Estimation

Not all measurements are done with instruments whose error can be reliably estimated. A classic example is the measuring of time intervals using a stopwatch. Of course, there will be a read-off error as discussed in the previous sections. However, that error will be negligible compared to the dominant error, the one coming from the fact that we, human beings, serve as the main measuring device in this case. Our individual reaction time in starting and stopping the watch will by far be the major source of imprecision. Since humans do not have built-in digital displays or markings, how do we estimate this dominant error?

The solution to this problem is to repeat the measurement many times. Then the average of our measurements is likely to be more closer to the true value than a single measurement. For instance, suppose you measure the oscillation period of a pendulum five times using a stopwatch.



Trial number, i	1	2	3	4	5
Measured Value, x_i (seconds)	3.9	3.5	3.7	3.4	3.5
Deviation, $d_i = x_i - \bar{x}$ (seconds)	0.3	-0.1	0.1	-0.2	-0.1

Our best estimate for the oscillation period is the average of the five measured values:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^5 x_i = \frac{3.9 + 3.5 + 3.7 + 3.4 + 3.5}{5} = 3.6 \text{ s}$$

Note that n in the general formula stands for the number of values you average.

Now, what is the error of our measurement? One possibility is to take the difference between the most extreme value and the average. In our case the maximum deviation is $(3.9 \text{ s} - 3.6 \text{ s}) = 0.3 \text{ s}$. If we quote 0.3 s as the error, we can be very confident that if we repeat the measurement again, we will find a value within this error of our average result.

The trouble with this method is that it overestimates the error. After all, we are not interested in the maximum deviation from our best estimate. We are much more interested in the average deviation from our best estimate. So, should we just average the differences from our measured values to our best estimate? Let us try:

$$\bar{d} = \frac{1}{n} \sum d_i = \frac{(0.3) + (-0.1) + (0.1) + (-0.2) + (-0.1)}{5} = 0.0 \text{ s}$$

Clearly, the average of deviations cannot be used as the estimation of the error since it gives us zero. In fact, the definition of the average ensures that the average deviation is always zero for any set of measurements. It is so because the deviations with positive sign are always canceled by the deviations with negative sign. Can we get rid of the negative signs? We can. If we square our deviations, all numbers will be positive, so we will never get zero. Then we should not forget to take the square root since our error should have the same units as our measured value.

Thus, we arrive at the famous standard deviation formula.

$$\sigma_x = \sqrt{\frac{1}{(n-1)} \sum_{i=1}^n (x_i - \bar{x})^2} = \sqrt{\frac{1}{(n-1)} \sum_{i=1}^n d_i^2} = \sqrt{\frac{0.09 + 0.01 + 0.01 + 0.04 + 0.01}{4}} = 0.2 \text{ s}$$

The standard deviation does not tell us exactly what we were looking for. It tells us what the average spread of experimental results is about the mean value. This is the error associated with a single measurement, among n measurements.

The error associated with the mean value of several measurements is the standard error.

$$\Delta x = \sigma_x / \sqrt{n} = \sqrt{\frac{1}{n(n-1)} \sum (x_i - \bar{x})^2} = \sqrt{\frac{1}{n(n-1)} \sum d_i^2} = \sqrt{\frac{0.09+0.01+0.01+0.04+0.01}{5 \times 4}} = 0.089 \text{ s}$$

This is called the standard error of the mean.

Error of a Measurement

Depending on the measuring tool and the technique used the estimated error of a measurement can be smaller or larger than the limiting error (least count) of the instrument. When stating the error, we always report the maximum possible error of a measurement.

- ❖ If the statistically estimated error, Δx is less than the least count of the measuring tool, then the error associated with the measurement is taken to be the least count.
- ❖ If Δx is greater than the least count of the measuring tool, then the error associated with the measurement is taken to be the estimated error Δx .

Example: The following measurements were taken using a micrometer screw gauge to find the diameter of a long thin wire.

$d_i(\text{mm})$	1.41	1.40	1.42	1.43	1.43	1.40	1.40	1.42
$(d_i - \bar{d})^2 \times 10^{-2}$	0.00	0.01	0.01	0.04	0.04	0.01	0.01	0.01

Mean value of the diameter = 1.413 mm = 1.41 mm

$$\Delta d = 0.005 < 0.01 \text{ mm (least count)}$$

$$\text{Diameter} = (1.41 \pm 0.01) \times 10^{-3} \text{ m}$$

Propagation of Errors

Often, we have two or more measured quantities that we combine arithmetically to get some result. Examples include dividing a certain distance by a certain time to get a speed or adding two lengths to get total length. Now that we have learned how to determine the error in the directly measured quantities, we need to learn how these errors propagate to an error in the result.

We assume that the two directly measured quantities are x and y , with errors Δx and Δy respectively. The measurements x and y must be independent of each other.

Fractional Error

The fractional error is the value of the error divided by the value of the quantity:

$$\text{The fractional error of } x = \frac{\Delta x}{x}.$$

The fractional error multiplied by 100 is the percentage error.

$$\text{The percentage error of } x = \frac{\Delta x}{x} \times 100\%$$

Everything in this section assumes that the error is "small" compared to the value itself, i.e., that the fractional error is much less than one.

For many situations, we can find the error in the result Z using three simple rules:

Rule 1: Constant multiplication

$$\text{If } Z = C \times x, \text{ where } C \text{ is a constant, then the error in } Z, \quad \Delta Z = C \times \Delta x$$

Rule 2: Addition or Subtraction

$$\text{If } Z = x \pm y \quad \text{then the error in } Z, \quad \Delta Z = \sqrt{(\Delta x)^2 + (\Delta y)^2}.$$

In words, this says that the error in the result of an addition or subtraction is the square root of the sum of the squares of the errors in the quantities being added or subtracted. This mathematical procedure, also used in Pythagoras' theorem about right triangles, is called quadrature.

Rule 3: Multiplication or Division

If $Z = x \times y$ or $Z = \frac{x}{y}$ then the fractional error in Z ,
$$\frac{\Delta Z}{Z} = \sqrt{\left(\frac{\Delta x}{x}\right)^2 + \left(\frac{\Delta y}{y}\right)^2}$$

In this case also the errors are combined in quadrature, but this time it is the fractional errors, i.e., the error in the quantity divided by the value of the quantity, that are combined. Sometimes the fractional error is called the relative error.

Then the error in Z ,
$$\Delta Z = Z \times \sqrt{\left(\frac{\Delta x}{x}\right)^2 + \left(\frac{\Delta y}{y}\right)^2}$$

Rule 4: Error in powers

If $Z = x^n$ then the fractional error in Z ,
$$\frac{\Delta Z}{Z} = n \times \frac{\Delta x}{x}$$

In general, for any function $Z = f(x, y, z, \dots)$, the error of Z can be estimated using

$$\Delta Z = \sqrt{\left\{\left(\frac{\partial f}{\partial x}\right) \Delta x\right\}^2 + \left\{\left(\frac{\partial f}{\partial y}\right) \Delta y\right\}^2 + \left\{\left(\frac{\partial f}{\partial z}\right) \Delta z\right\}^2 + \dots}$$

Example: Consider $Z = \ln x$

then the error
$$\Delta Z = \left(\frac{\partial \ln x}{\partial x}\right) \Delta x = \Delta x / x.$$

Least Squares Error Analysis for Linear Graphs

Finding the gradient and intercept of a linear graph is relatively straight forward. But precision is lost during the process of plotting the graph and extracting the slope and intercept. Therefore, in this method, two people evaluating the same data would get different results.

The usual scientific approach used when a curve is fitted to a set of data is the least squares analysis. This analysis can be used for different types of graphs. But the case of fitting a straight line to data is presented here. For this, the following assumptions are used.

- 1) The uncertainties are the same for all of the data points.
- 2) The uncertainties are only in the dependent variable, y .
- 3) The uncertainties are all random and the multiple measurements for a given value of the independent variable, x , would be distributed according to a Gaussian distribution.

Consider the data to be in pairs $(x_i \text{ and } y_i)$. The goal is to find the straight line ($y = mx + c$) that best fit the given data and to determine the errors associated.

Then, the following statistical equations for the gradient and the intercept can be derived using the least square analysis.

$$m = \frac{1}{\Delta} \left[n \left(\sum_{i=1}^n x_i y_i \right) - \left(\sum_{i=1}^n x_i \right) \left(\sum_{i=1}^n y_i \right) \right]$$

$$c = \frac{1}{\Delta} \left[\left(\sum_{i=1}^n y_i \right) \left(\sum_{i=1}^n x_i^2 \right) - \left(\sum_{i=1}^n x_i \right) \left(\sum_{i=1}^n x_i y_i \right) \right]$$

where

$$\Delta = n \left(\sum_{i=1}^n x_i^2 \right) - \left(\sum_{i=1}^n x_i \right)^2.$$

And the following are the statistical equations for errors associated with gradient (m) and the intercept (c) in terms of the error associated with y coordinates of the graph.

$$\Delta m = \Delta y \sqrt{\frac{n}{\Delta}} \quad \text{and} \quad \Delta c = \Delta y \sqrt{\frac{1}{\Delta} \left(\sum_{i=1}^n x_i^2 \right)}$$

where

$$\Delta y = \sqrt{\frac{1}{n} \left(\sum_{i=1}^n (y_i - mx_i - c)^2 \right)}.$$

As the Least Square Analysis described above is tedious to calculate, the following simplified method can be followed to determine the gradient, the intercept, and their errors of a linear graph.

Error Associated with the Gradient of a Graph (Simple Method)

Following are the simplified steps to be followed to calculate the gradient, the intercept, and their errors of a linear graph.

1. Find the average value of x -coordinates ($\bar{x}$) and y -coordinates ($\bar{y}$) of the points that are used to plot the graph.

$$\bar{x} = \frac{1}{n} \left(\sum_{i=1}^n x_i \right) \quad \bar{y} = \frac{1}{n} \left(\sum_{i=1}^n y_i \right)$$

where x_i and y_i are coordinates of the data points and n is the number of data points so that $i = 1, 2, \dots, n$.

2. Plot the data points and draw the best fit line so that it passes through the point $(\bar{x}, \bar{y})$.
3. Calculate the gradient (m) and intercept (c) of the best fit line passing through the point $(\bar{x}, \bar{y})$
4. Calculate gradients (m_i) and intercepts (c_i) using each data point using the following equations.

$$m_i = \frac{y_i - \bar{y}}{x_i - \bar{x}} \quad \& \quad c_i = y_i - mx_i$$

where $i = 1, 2, \dots, n$.

5. Determine the errors associated with the gradient (Δm) and intercept (Δc) of the best fit line using the following equations.

$$\Delta m = \sqrt{\frac{1}{n(n-1)} \sum (m_i - m)^2} \quad \& \quad \Delta c = \sqrt{\frac{1}{n(n-1)} \sum (c_i - c)^2}$$

Stating the Error and the Result

The result of a measurement (x) can be stated as $x = x_{\text{exp}} \pm \Delta x$.

When stating the result, there are two rules to follow.

1. **Error should never be quoted with more than two significant figures.**
2. **The result should be quoted to the same number of decimal places as in the error.**

Example:

Consider the error and the result of a quantity is found as (say) 0.01325 J/cal and 4.1864 J/cal respectively.

The result should be given as 4.186 ± 0.013 J/cal.

Since error can be stated with either one or two significant figures, as a practice it is better choosing the number of significant figures in the error consistently. Otherwise, one can state the error with two significant figures while the other stating the same error with one significant figure. Therefore, let us use the following method to decide the number of significant figures in the error.

If the first significant figure of the error is “1” (one) after the error is rounded to two significant figures, then the error should be stated by two significant figures. Otherwise, the error can be stated with only one significant figure.

For example, consider the uncertainty of two different measurements to be 0.016 and 0.34 after rounding off to two significant figures. Then the error of these measurements should be stated as 0.016 and 0.3, respectively.

If the error of the result in the above example was found to be 0.02714 J, then the result should be given as 4.19 ± 0.03 J.